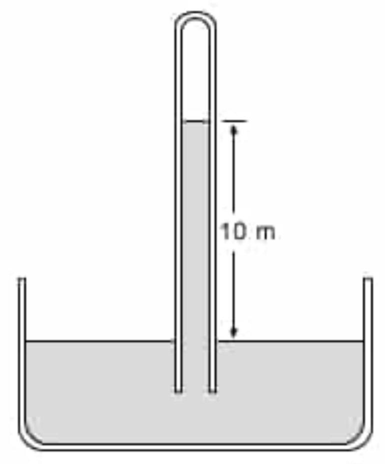
Atmospheric pressure on a distant planet causes water to rise to a height of 10 m inside an evacuated tube with its open end immersed in a container of water open to ambient pressure.

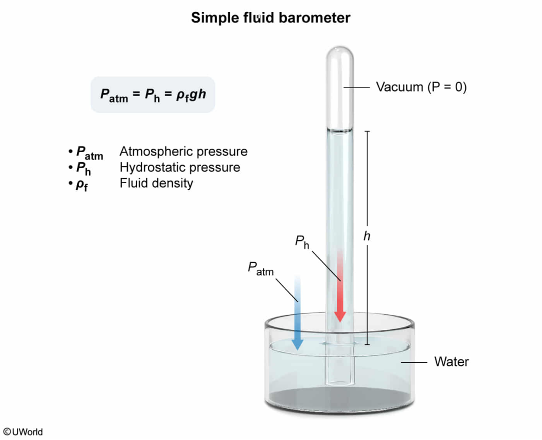


What is the atmospheric pressure? (Note: The density of water is 10^3 kg/m^3 and the gravitational acceleration is 20 m/s^2)

- A. 2×10^4 N/m^2 [12%]
- B. 2×10^5 N/m^2 [74%]
- C. 2×10^6 N/m^2 [11%]
- D. 2×10^7 N/m^2 [1%]

Correct
74% answered correctly

Explanation:



Pressure is a force exerted perpendicular to an area. **Atmospheric pressure** P_{atm} refers to the pressure that gas molecules in the atmosphere exert on all terrestrial objects (eg, bodies of water, living beings).

A **simple fluid barometer** is a device that may be used to measure P_{atm}. Unlike gases in the

atmosphere, fluids occupying the evacuated tube of a simple fluid barometer are considered ideal and therefore do not expand or contract in response to local conditions such as weather and elevation. As a result, the hydrostatic pressure P_h generated by the fluid column inside of the barometer is a reliable estimate of local P_{atm}:

$$P_{\text{atm}} = P_{\text{h}}$$

P_h itself is related to the fluid density ρ_f, gravitational acceleration g, and total fluid column height h. Therefore, P_{atm} may be quantified in terms of the variables influencing P_h:

$$P_{\text{atm}} = P_{\text{h}} = \rho_{\text{f}}\text{gh}$$

In this question, the P_{atm} of the distant planet (g = 20 m/s^2) causes a column of water (ρ = 10^3 kg/m^3) to rise to a height of 10 m in an evacuated tube. Applying these values to the equation above yields:

$$P_{\text{atm}} = \left(10^3\ \frac{\text{kg}}{\text{m}^3}\right)\left(20\ \frac{\text{m}}{\text{s}^2}\right)(10\ \text{m})$$

$$P_{\text{atm}} = 2\times10^5\ \frac{\text{kg}\cdot\text{m}^2}{\text{m}^3\cdot\text{s}^2} = 2\times10^5\ \frac{\text{kg}\cdot\text{m}}{\text{m}^2\cdot\text{s}^2}$$

Reorganizing the units verifies that pascals (Pa), a common unit of pressure, are obtained:

$$P_{\text{atm}} = \left(2\times10^5\ \frac{\text{kg}\cdot\text{m}}{\text{s}^2}\cdot\frac{1}{\text{m}^2}\right) = \left(2\times10^5\ \text{N}\cdot\frac{1}{\text{m}^2}\right) = 2\times10^5\ \text{Pa}$$

Therefore, it can be concluded that the P_{atm} of the distant planet is **2 × 10^5 N/m^2 (Choice B)**, which is equivalent to 2 × 10^5 Pa (1 N/m^2 = 1 Pa).

(Choice A) 2 × 10^4 N/m^2 is obtained if 1 m instead of 10 m is used for the height of the water inside the tube.

(Choices C and D) 2 × 10^6 N/m^2 or 2 × 10^7 N/m^2 is obtained if the incorrect value of 100 kg/m^3 or 10 kg/m^3, respectively, is used for the density of water.

Educational objective:
A simple fluid barometer may be used to measure atmospheric pressure, the pressure exerted on all terrestrial objects by gas molecules in the atmosphere. Atmospheric pressure is equal to the hydrostatic pressure fluid column of a simple fluid barometer.

Time spent:0 sec
Subject:
Physics

QID:402146
System:
Fluids